

Machine Learning

MARKOV CHAIN

IT

ML

Markov chain

A **Markov Chain** is a mathematical model used to describe a system that **moves from one state to another**, where the next state depends **only on the current state**, not on the entire past history

The future depends only on the present, not on the past .

The main rule is:

$$P(X_{n+1} = x_{n+1} | X_n = x_n, X_{n-1} = x_{n-1}, \dots, X_0 = x_0)$$

is equal to

$$P(X_{n+1} = x_{n+1} | X_n = x_n)$$

Question

A company observes customer behavior between two states:

- State S_1 : Buy Product
- State S_2 : Do Not Buy

If a customer buys the product today, find the probability that the customer buys the product after 2 days.

Transition Probabilities

Current State	Buy (S_1)	Not Buy (S_2)
Buy (S_1)	0.7	0.3
Not Buy (S_2)	0.4	0.6

Given

If the customer buys today (S_1)

Probability of buying again tomorrow = 0.7

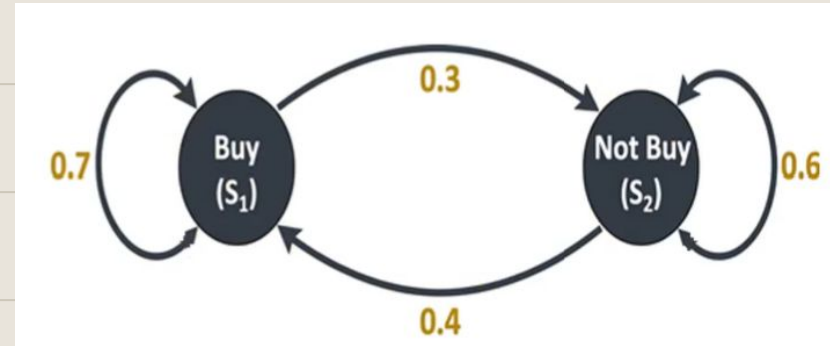
Probability of not buying tomorrow = 0.3

If the customer does not buy today (S_2)

Probability of buying tomorrow = 0.4

Probability of not buying tomorrow = 0.6

State Diagram



Solution

The Transition matrix is given as

$$P = \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix}$$

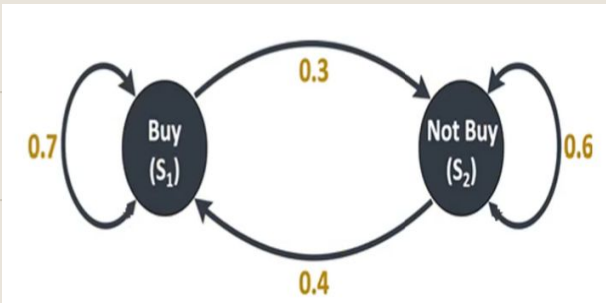
In Markov Chains, the probability after n steps is obtained by multiplying the transition matrix n times.

So, for 2 days:

$$p_2 = P \times P$$

This gives the probabilities of moving from one state to another in two steps.

Current State	Buy (S_1)	Not Buy (S_2)
Buy (S_1)	0.7	0.3
Not Buy (S_2)	0.4	0.6



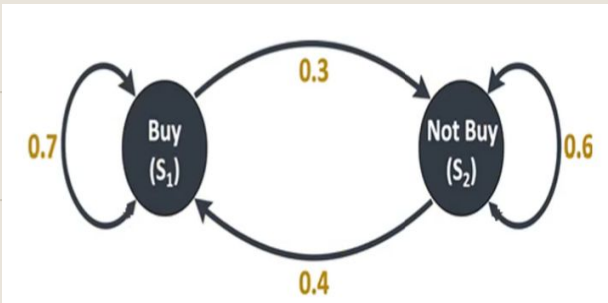
Solution

The Transition matrix is given as

$$P2 = \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix} \times \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix}$$

$$P2 = \begin{bmatrix} 0.61 & 0.39 \\ 0.52 & 0.48 \end{bmatrix}$$

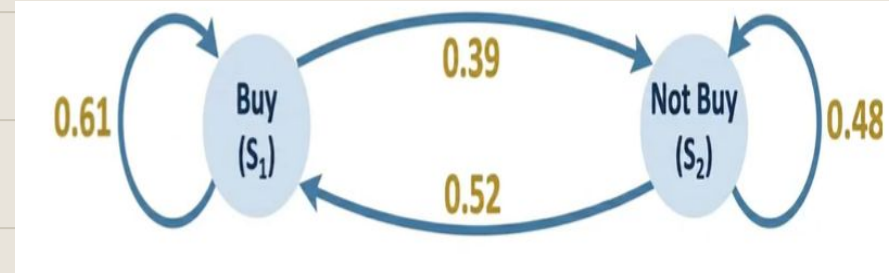
Current State	Buy (S_1)	Not Buy (S_2)
Buy (S_1)	0.7	0.3
Not Buy (S_2)	0.4	0.6



Solution

The Transition matrix is given as

$$P^2 = \begin{bmatrix} 0.61 & 0.39 \\ 0.52 & 0.48 \end{bmatrix}$$



If the system starts in state S₁ (Buy)

Probability of Buying after 2 steps = 0.61.